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An Attention-Based CNN for ECG Classification

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Abstract. The paper considers the problem of improving the interpretability of a convolutional neural network on the example of ECG classification task. This is done by using an architecture based on attention modules. Each module generates a mask that selects only those features that are required to make the final prediction. By visualizing these masks, areas of the signal that are important for decision-making can be identified. The model was trained both on raw signals and on their logarithmic spectrograms. In the case of raw signals, generated masks did not perform any meaningful feature maps filtering, but in the case of spectrograms, interpretable masks responsible for noise reduction and arrhythmic parts detection were obtained.

Keywords: convolutional neural networks, attention mechanism, ECG classification

1 Introduction

In the last few years, deep neural networks have been successfully applied to a large variety of tasks achieving state-of-the-art results. But in the fields where the cost of a mistake is high, such as medical diagnostics, model performance is not the only thing that matters: the person who makes the final decision needs to understand the model's behavior and particular features of input data that affected its prediction. Thus, it is crucial to obtain not only an accurate model but also an interpretable one.

The paper is focused on building such a model for an ECG classification task. To accomplish this, a convolutional neural network architecture based on an attention mechanism was used. This mechanism allows discovering the most informative features of an ECG signal at various network depths. Models with different forms of attention have already been successfully applied to a large variety of tasks including neural machine translation, speech recognition, caption generation and image classification. The model was trained both on raw signals and on their logarithmic spectrograms. As a result, interpretable masks responsible for noise reduction and arrhythmic parts detection were obtained.

2 Datasets

Two datasets were used for model training and testing: publicly available part of the 2017 PhysioNet/CinC Challenge dataset [1, 2] and the MIT-BIH Atrial Fibrillation Database [3].

The PhysioNet dataset contains 8,528 single-lead ECG recordings lasting from 9 to 61 seconds with an equal sampling rate of 300 Hz. All ECGs were collected from portable heart monitoring devices. A significant part of the signals had their R-peaks directed downwards since the device did not require the user to hold it in any specific orientation. These signals were flipped during the preprocessing stage. All recordings were manually classified by a team of experts into 4 classes: atrial fibrillation, normal rhythm, other rhythm or too noisy to be classified. For this dataset, the model was trained to predict the correct class.

The MIT-BIH Atrial Fibrillation Database contains 23 approximately 10-hour ECG recordings. Each signal was sampled at 250 Hz and has two leads, but we used only the first one for model training. The signals are divided into segments marked with one of four labels: atrial fibrillation, atrial flutter, junctional rhythm or other rhythms. Such annotation allows sampling segments around the points of heart rhythm change and define model's target as a fraction of an arrhythmic part in the entire segment.

3 Model

The model is based on Residual Attention Network [4] and is constructed by stacking multiple ResNet blocks [5] and slightly modified attention modules.

3.1 Attention module

Each attention module is a neural network that starts with a pre-processing ResNet block, then splits into a trunk branch and a mask branch, that are then joined together and passed through a post-processing ResNet block. The module's architecture is shown in Figure 1.

The trunk branch performs feature processing by passing its input through two consecutive ResNet blocks. The mask branch creates a soft binary mask that highlights the most informative features of an input ECG signal. Leading and trailing convolutions in this branch are used to squeeze and expand the number of feature maps. Upsampling is performed by linear interpolation. The computed mask is subsequently multiplied by the output of the trunk branch. In order to help the gradient flow and preserve good properties of features, the output of the trunk is then added to the result of this multiplication as suggested in [4].

3.2 Model architecture

The model trained on raw signals has two leading ResNet blocks with 16 filters, three attention modules with 20 filters and finally 2 trailing ResNet blocks

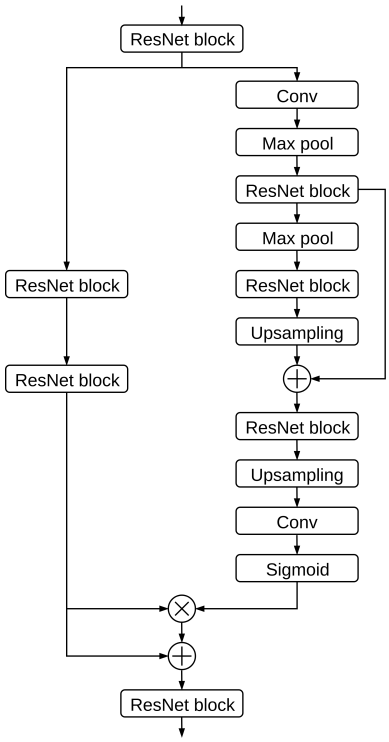


Fig. 1: Attention module architecture. Trunk branch is shown on the left, mask branch – on the right.

with 24 filters. The model trained on logarithmic spectrograms consists of two attention modules and a trailing ResNet block with 12 filters each. All of these attention modules and top-level ResNet blocks are followed by a dropout operation [6] with a drop rate of 0.2 and perform downsampling along spatial dimensions. In a ResNet block, the downsampling is performed after the first convolution using a max pooling operation with a kernel size and a stride of 2. In an attention module, the downsampling is performed by the post-processing ResNet block.

In both models, first convolutions in mask branches squeeze the number of feature maps to 4, while the last convolution expands them back to match the shape of the corresponding trunk branch. These convolutions have a kernel size of 1, all others have a kernel size of 5. The convolution stride in all layers is fixed to 1. The input of each convolutional layer is padded in such a way that the spatial resolution is preserved afterwards. ReLU activations [7] are used throughout both networks, batch normalization [8] is used before each activation.

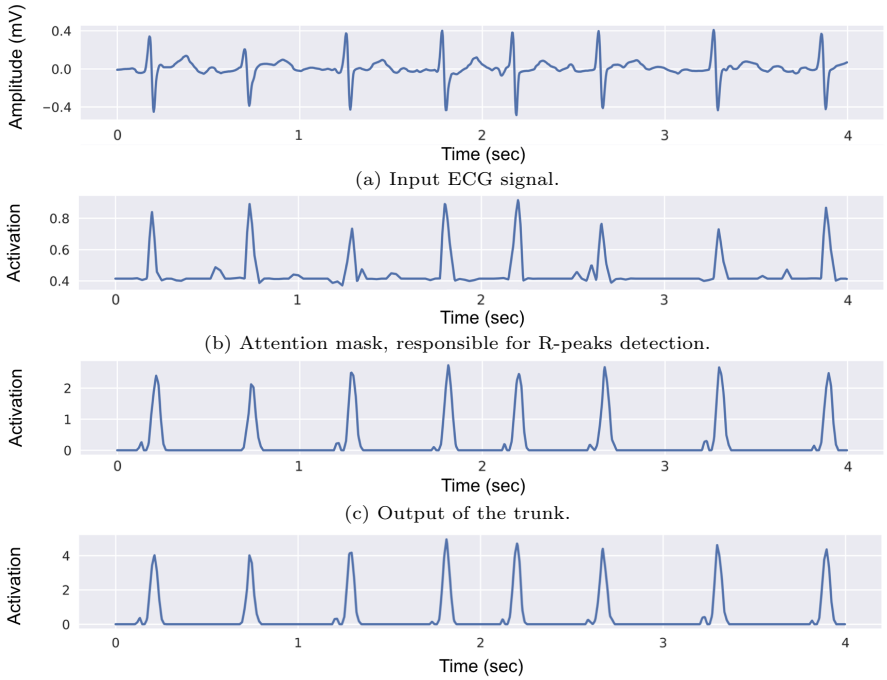
Both networks end with a global max pooling operation and a fully-connected layer. In the case of PhysioNet dataset, this layer is followed by a softmax ac-

tivation to predict class probabilities and categorial cross entropy is used as a loss function. In the case of MIT-BIH Atrial Fibrillation Database, sigmoid activation and binary cross entropy loss are used. All weights were initialized by the scheme, proposed in [9], adam [10] was used as an optimizer. Described models were trained on 10-seconds crops from original signals with the help of (anonymized) framework for ECG processing [11].

4 Results

4.1 Raw signals

Figure 2 shows one signal from the PhysioNet dataset and a predicted mask, responsible for R-peaks detection. The problem lies in the fact that the trunk branch is able to detect the peaks by its own. Almost all of the meaningful attention masks behave this way: they highlight only those areas, where the corresponding feature map from the trunk branch activates. This means, that mask branches don't perform feature selection, but only scale the output of the trunk.



(a) Input ECG signal.

(b) Attention mask, responsible for R-peaks detection.

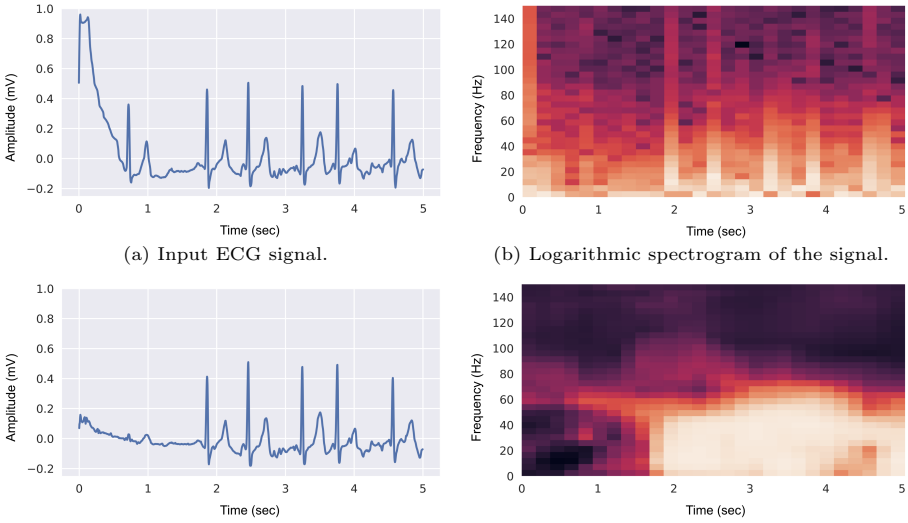
(c) Output of the trunk.

(d) Input to the post-processing ResNet block.

Fig. 2: An example of a mask, responsible for R-peaks detection.

4.2 Logarithmic spectrograms

More useful and interpretable masks can be obtained by switching from raw signals to their logarithmic spectrograms. Figure 3 illustrates one of the masks predicted by the network, trained on the PhysioNet dataset. In the first two seconds, it suppresses almost all the frequencies in the recording, while in the rest of the signal, it acts as a low-pass filter with a cutoff frequency of about 60 Hz. This mask can be applied to the original signal by taking its short-time Fourier transform, multiplying it by the mask, and performing the inverse transform. This operation results in noise removal at the beginning of the recording.



(a) Input ECG signal.

(b) Logarithmic spectrogram of the signal.

(c) Input signal with the mask applied.

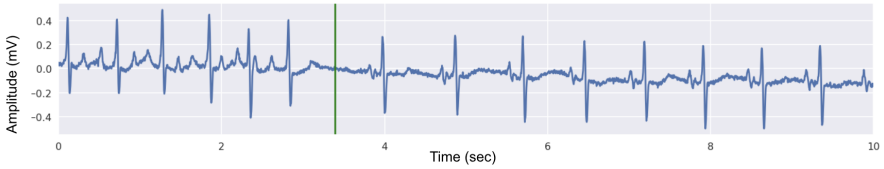
(d) Predicted mask.

Fig. 3: An example of a mask, responsible for noise reduction.

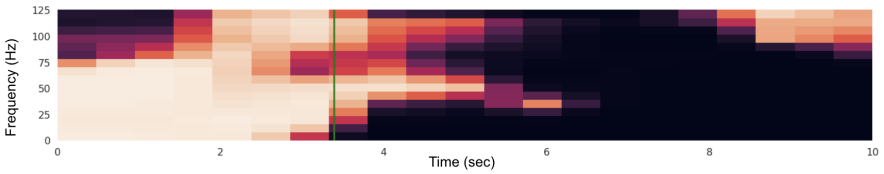
Figure 4 shows an ECG from the MIT-BIH Atrial Fibrillation Database and two masks responsible for the detection of arrhythmic and non-arrhythmic parts of the signal. Recall that in this case, the model did not know the position of the boundary of heart rhythm change, but only a fraction of an arrhythmic part in the entire segment.

5 Conclusion

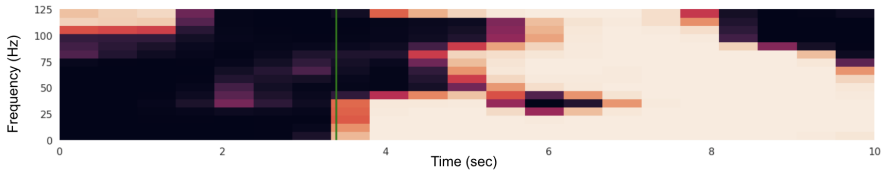
In this work, an interpretable convolutional neural network based on the attention mechanism was considered on the example of ECG classification task. Such architecture allows a better understanding of the model's decision-making process by visualizing masks generated by attention modules. The model was trained both on raw signals and on their logarithmic spectrograms. In the first



(a) Input ECG signal.



(b) Attention mask, responsible for arrhythmic segment detection.



(c) Attention mask, responsible for non-arrhythmic segment detection.

Fig. 4: An example of masks, responsible for arrhythmic and non-arrhythmic segments detection. The green line represents the boundary of heart rhythm change.

case, generated masks did not perform any meaningful feature maps filtering, but in the second case, interpretable masks responsible for noise reduction and arrhythmic parts detection were obtained. This means that the attention mechanism behaves differently depending on the type of input data and it is important to determine whether the predicted masks actually perform feature selection.

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